

[A.]

what is the hydraulic gradient? 1.25

darcy flux? $1.25 \times 0.5 = \frac{5}{8} = 0.625$

volume after 2 days? $\pi \cdot \left(\frac{1.30}{2}\right)^2 \cdot 0.625$

$$\pi \cdot 0.4225$$

$$1.327 \cdot 0.625$$

$$0.8295 \text{ M}^3/\text{day}$$

$$[0.8295 \times 2] = \underline{1.66 \text{ M}^3}$$

$$\frac{2.5}{2}$$

B. 1) $\frac{220 - 130}{250} = \frac{90}{250} = \frac{dh}{dl} = 0.36$

2) Darcy flux = $0.36 \times 0.02 = 0.0072 \text{ m/day}$
(specific discharge)

3) $\left[\pi \left(\frac{1.4}{2} \right)^2 \cdot 0.0072 \right] \cdot 2$

$$2 \left[\pi (0.49) \cdot 0.0072 \right]$$

$$2 \cdot 1.539 \cdot 0.0072$$

$$0.02 \text{ M}^3 \text{ discharge}$$